

Berry curvature and response functions from VASP wavefunctions

VASPBERRY 1.3 scientific examples

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Abstract

VASPBERRY evaluates geometric and response properties from VASP electronic states. This report illustrates the discrete-wavefunction and Kubo approaches with monolayer MoS₂ and a Bi bilayer. The examples cover reciprocal-space Berry curvature, Chern and Z₂ indices, intrinsic charge Hall response, circular optical transitions and real-space spinor densities. Berry-curvature maps use Cartesian coordinates, band paths use distances along named high-symmetry directions, and the Z₂ n-field is shown in reduced coordinates to display its half-zone sums. The calculation guides provide the VASP files and executable commands.

1. Methods

1.1 Berry curvature from wavefunction overlaps

The Fukui–Hatsugai–Suzuki method evaluates the Berry phase around each cell of a periodic $\mathbf{k}$ mesh using overlaps between neighboring wavefunctions. For an isolated group of bands, the overlap is a matrix and its determinant gives a gauge-invariant loop phase. With the connection $\mathbf{A} = i\langle u | \nabla_{\mathbf{k}} u \rangle$ used here,

$$\Phi_p = -\arg(U_1(\mathbf{k})U_2(\mathbf{k} + \Delta\mathbf{k}_1)U_1(\mathbf{k} + \Delta\mathbf{k}_2)^{-1}U_2(\mathbf{k})^{-1}), \quad \overline{\Omega}_p = \frac{\Phi_p}{\Delta S_k}.$$

The normalized link U_j is the phase of the overlap determinant. $\Delta S_k = |\mathbf{b}_1 \times \mathbf{b}_2|/(N_1 N_2)$ is the area of a mesh cell, with reciprocal vectors containing 2π . Thus Φ_p is a phase and $\overline{\Omega}_p$ is an area-averaged curvature in Å². The Chern number is $C = (2\pi)^{-1} \sum_p \Phi_p$. An isolated occupied group can include internal degeneracies; its separation from excluded bands remains essential. The discrete construction and its band-group extension are described by Fukui, Hatsugai and Suzuki.

The map and integral convey different information. A material can have strong local curvature at opposite valleys while its total Chern number is zero. An integer lattice Chern result also needs a sufficiently resolved mesh and a physically appropriate isolated band group.

1.2 Kubo curvature and Hall conductivity

For an isolated band and a Cartesian Hamiltonian derivative $D_\alpha = \partial H / \partial k_\alpha$,

$$\Omega_{n,xy}(\mathbf{k}) = -2 \text{Im} \sum_{m \neq n} \frac{D_{x,nm} D_{y,mn}}{(E_n - E_m)^2}.$$

This form resolves curvature at each sampled $\mathbf{k}$ point. Sharp curvature near small gaps makes the $\mathbf{k}$ sampling and intermediate-band window particularly important. Individual-band values are not used where the sampled bands cannot be resolved separately. The native WAVECAR implementation uses canonical momentum of the stored pseudo-wavefunctions; full material velocity can require PAW, nonlocal and SOC terms. The exported-matrix interface permits an explicitly supplied physical operator.

For an isolated occupied bundle $\mathcal{V}$, the trace curvature is

$$\Omega_{xy}^{\mathcal{V}} = -2 \text{Im} \sum_{n \in \mathcal{V}} \sum_{m \notin \mathcal{V}} \frac{D_{x,nm} D_{y,mn}}{(E_n - E_m)^2}.$$

Internal occupied-pair terms cancel in the sum of band curvatures. Excluding them before division avoids singular individual-band terms at internal degeneracies. The bundle must remain separated from the excluded bands. This occupied-to-unoccupied formulation and its numerical advantage are given by Wang et al., Eq. (11) and Sec. III D.

For a two-dimensional system, the intrinsic charge sheet conductivity is

$$\sigma_{xy}(\mu, T) = -\frac{e^2}{\hbar} \sum_n \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} f(E_{n\mathbf{k}} - \mu, T) \Omega_{n,xy}(\mathbf{k}).$$

Within an insulating gap at zero temperature, this becomes $\sigma_{xy} = -(e^2/h)C_{\text{occ}}$ in the stated convention. General background on Berry curvature and electronic transport is given in Xiao, Chang and Niu. The Bi Hall example evaluates the occupied-group flux directly; it does not assign independent curvatures to unresolved Kramers partners.

2. Materials and numerical settings

Dataset	Wavefunctions and sampling	Quantities shown
Monolayer MoS ₂ , full zone	SOC; 12×12 mesh; 26 bands; 400 eV	Fukui and Kubo occupied-bundle curvature, bands 1–18
Monolayer MoS ₂ , matching path	SOC; 49 K–Γ–K′ points; 26 bands; 400 eV	Band structure and Kubo curvature beside the maps
Monolayer MoS ₂ , local valleys	SOC; two 9×9 patches; 26 bands; 400 eV	Isolated band-18 curvature near K and K′
Monolayer MoS ₂ , supplied path	SOC; 48 K–Γ–K′ points; 32 bands; 400 eV	Optical spectra and Γ-point state density
Buckled Bi bilayer	SOC; full 12×12 mesh; 18 bands; occupied bands 1–10	Fukui–Hatsugai n-field and Z_2 invariant

The input guide links the structures and VASP files. The MoS₂ maps and their matching path use the same public SCF charge density, structure, potentials, cutoff and 26-band window. VASP 5.4.4 generated the 12×12 mesh and 49-point path with ICHARG=11 and ISYM=−1; only the $\mathbf{k}$ -point list changes. The sampled occupied-to-empty direct gap is 1.674 eV. The tutorial supplies preparation files and commands for both WAVECARs. The optical and real-space examples retain the separate supplied 32-band path.

3. Results

3.1 MoS₂: Fukui Berry curvature over the full Brillouin zone

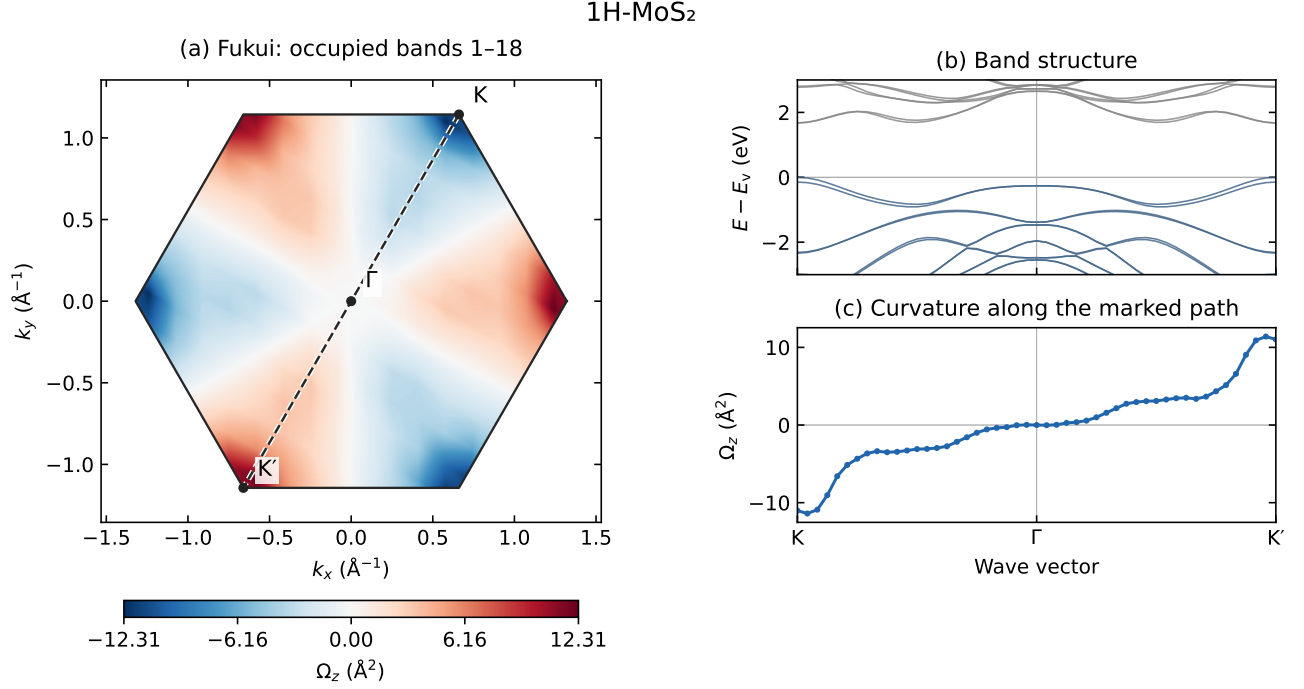


Figure 1. (a) Fukui Berry curvature of occupied bands 1–18 in the Cartesian first Brillouin zone. The dashed line marks K– Γ –K', with $K = (1/3, 2/3)$ and $K' = -K$ in reciprocal coordinates. Native 12×12 plaquette values are displayed with periodic bilinear interpolation onto a 401×401 Cartesian grid. (b) Band structure along that path; energies are relative to the valence-band maximum, with occupied bands in blue and empty bands in gray. (c) A periodic bilinear cut of the plaquette field along the marked path. This line visualizes panel (a), rather than adding an independent pointwise curvature calculation. Both right-hand panels share the same Cartesian path distance.

The curvature has opposite signs near the time-reversed K and K' valleys. The largest sampled magnitudes are approximately **12.31** Å², while the full-zone Chern number is zero to numerical precision. The native text map integrates to approximately -5×10^{-7} after decimal rounding. This illustrates the distinction between valley-contrasting local geometry and vanishing total charge Chern number. The connection between inversion breaking, spin–orbit coupling and valley optical response in MoS₂ is discussed by Xiao et al.. The present map is a finite-mesh example rather than a convergence study.

Calculation and plotting commands

3.2 MoS₂: Kubo curvature of the occupied valence-band bundle

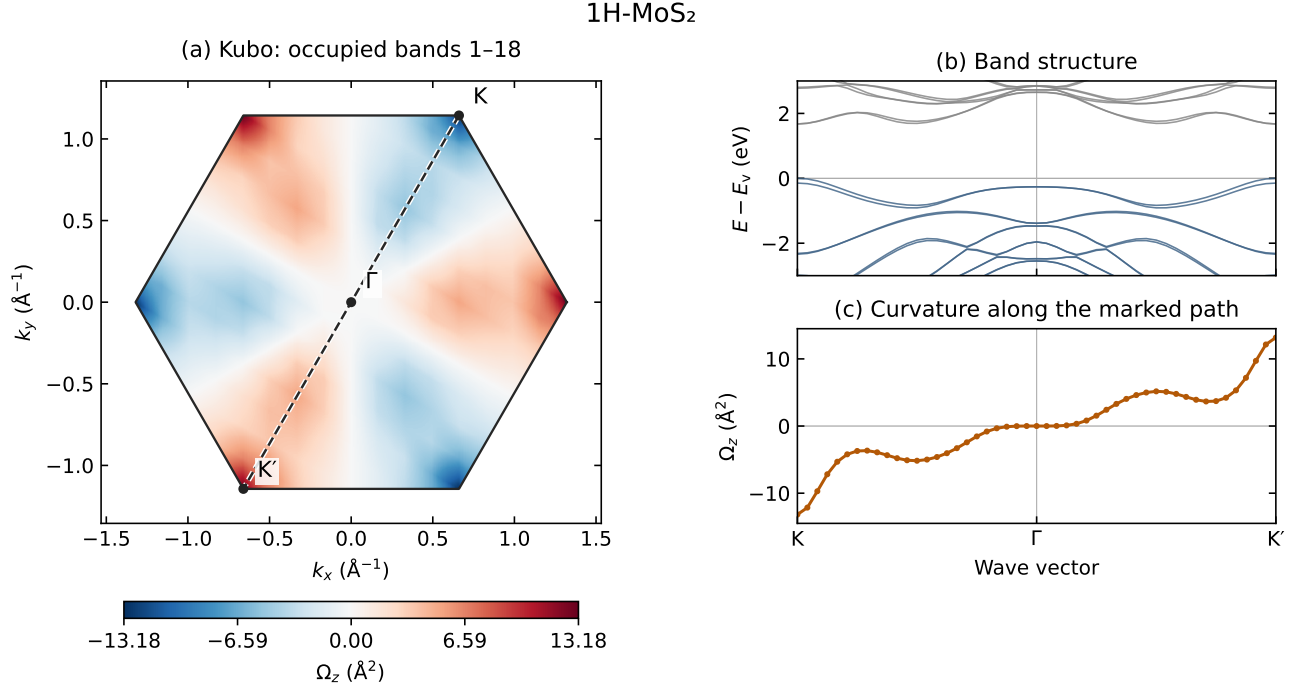


Figure 2. (a) Canonical-momentum Kubo trace curvature of occupied bands 1–18 on the 12×12 mesh. Only transitions to the stored empty bands 19–26 enter the sum. The map uses the same periodic bilinear display interpolation and dashed K – Γ – K' path as Figure 1. (b) Band structure, with occupied bands in blue and empty bands in gray. (c) Bundle curvature calculated directly at all 49 path points. Internal valence-band degeneracies do not interrupt the bundle curve.

The occupied-bundle curvature is approximately $-13.18 \text{ \AA}^2$ at K and $+13.18 \text{ \AA}^2$ at K' . The occupied space stays separated from the empty states by at least **1.674 eV** on both samplings, so every mesh and path point is valid for the bundle calculation. The native option `-kubo_bundle 1` performs the external-band sum directly, avoiding large cancelling internal terms.

Figures 1 and 2 now describe the same occupied band space. Their numerical values need not coincide at this resolution: Fukui gives finite-plaquette averages, while this Kubo result uses pointwise canonical momentum and a finite empty-band window. The 12×12 mesh and 26 stored bands require convergence for quantitative predictions. NumPy interpolation smooths only the display; all integration and numerical checks use the original samples.

Calculation and interpretation

3.2.1 MoS₂: an isolated single band near the valleys

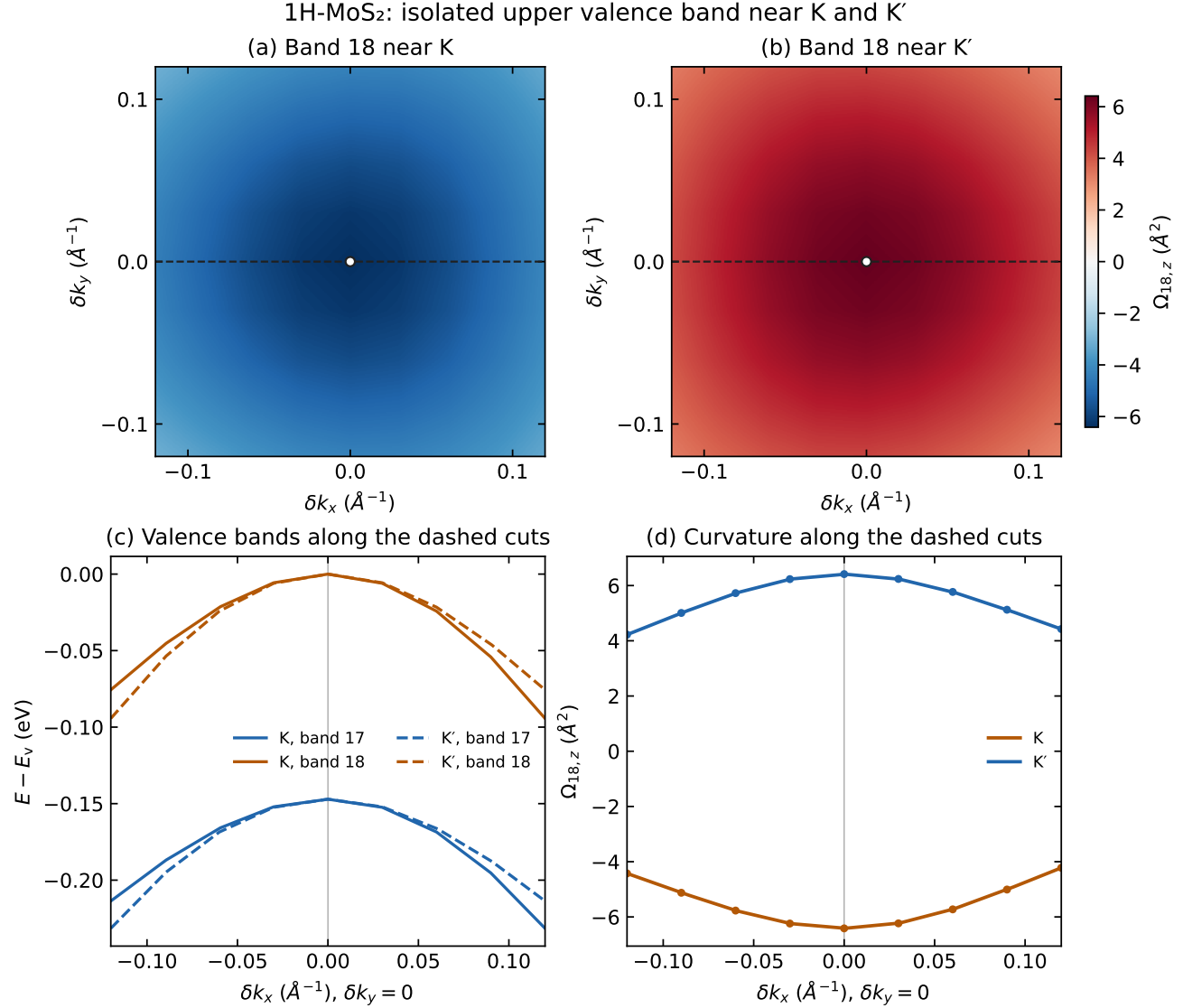


Figure 3. Single-band Kubo curvature of the upper valence band, band 18, around K and K'. Each local Cartesian patch extends $\pm 0.12 \text{ \AA}^{-1}$ and contains 9×9 VASP points. The upper maps use NumPy bilinear display interpolation; the dashed lines mark the cuts below. The lower panels show the two upper valence bands near both valleys and band-18 curvature at both valleys. Symbols mark the calculated curvature samples.

Band 18 touches its Kramers partner at Γ , but is separated from every other stored band by at least **0.130 eV** throughout these valley patches. The 17–18 splitting at K is **0.147 eV**, and the valley curvatures are approximately **$-6.415/+6.415 \text{ \AA}^2$** at **K/K'**. This is a suitable local single-band example of the spin-split valley physics described by Xiao et al.. The patch data describe local curvature; a full-zone integral requires the appropriate isolated band bundle and complete BZ sampling.

VASP preparation, calculation and reference data

3.3 MoS₂: valley-selective circular transitions

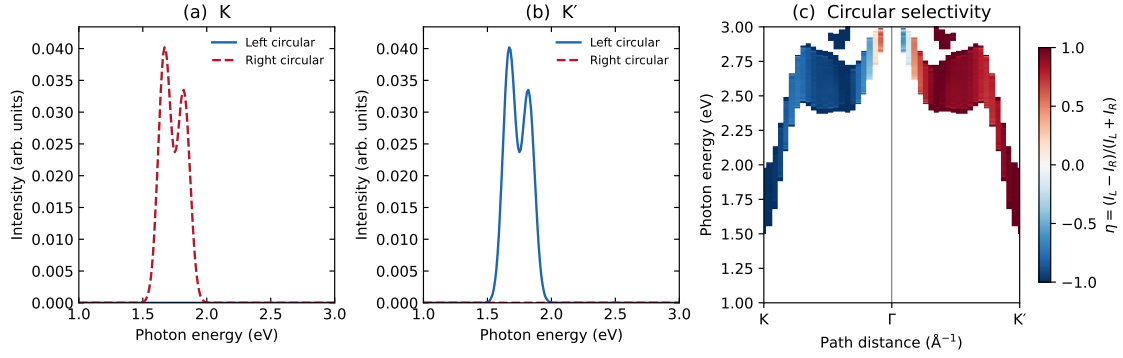


Figure 4. Left- and right-circular spectra at K and K' and the selectivity $\eta = (I_L - I_R)/(I_L + I_R)$ along the path. Normal incidence is used, with 0.05 eV Gaussian broadening and transitions from occupied bands 1–18 to bands 19–20. White regions in the selectivity map exclude negligible intensity.

The first peak occurs near **1.67 eV**. Its selectivity is -1 at K and $+1$ at K' in VASPBERRY's channel convention. The opposite helicities provide a clear illustration of valley-selective transitions. Intensities are in arbitrary units and retain the native sampling normalization; an absolute absorption spectrum requires a full-zone calculation and the appropriate optical matrix elements.

Optical calculation guide

3.4 Bi: Fukui–Hatsugai Z_2 invariant and n-field map

For the Bi bilayer, VASPBERRY evaluates the Fukui–Hatsugai lattice n-field from the occupied SOC bands 1–10 on a full Γ -centered 12×12 mesh. With the time-reversal-compatible gauge used in this construction, the Z_2 invariant is the parity of the integer-field sum over a half Brillouin zone:

$$\nu = \left[\sum_{p \in B_{1/2}} n(p) \right] \bmod 2.$$

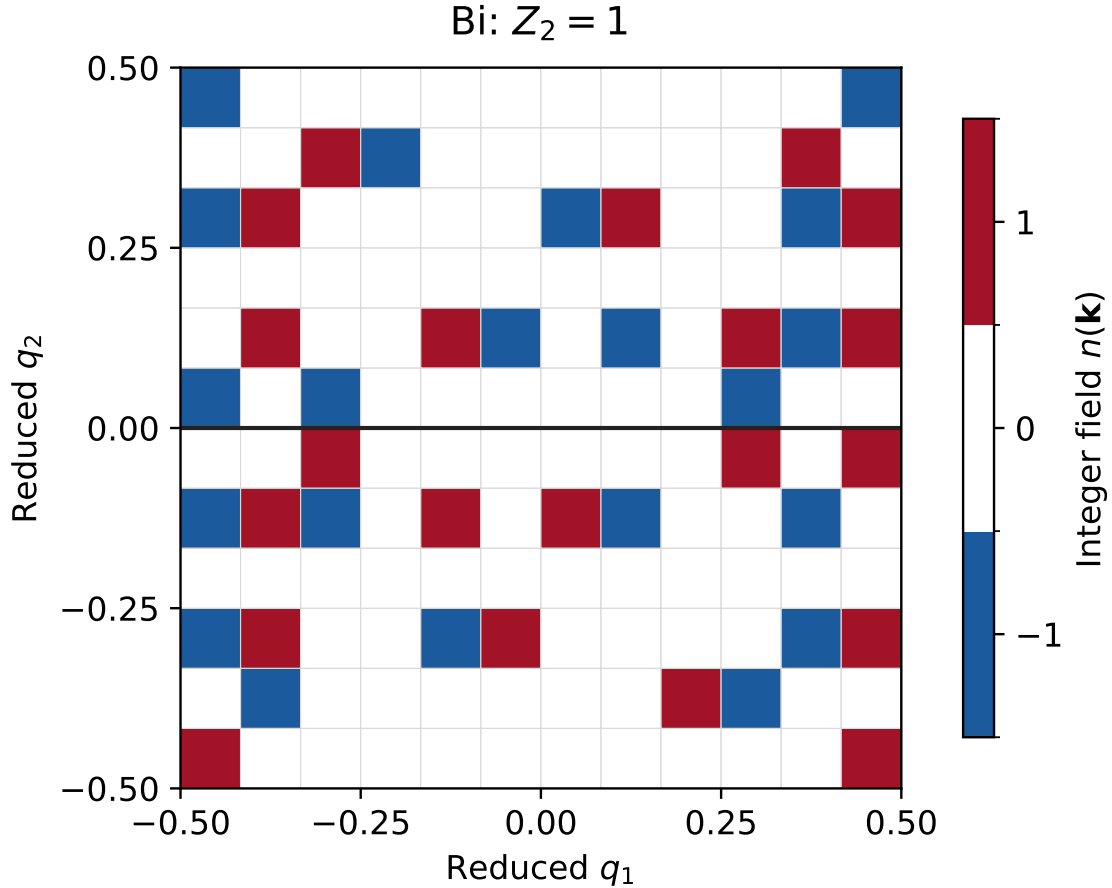


Figure 5. Fukui–Hatsugai integer field $n(\mathbf{k})$ calculated from the Bi VASP spinor wavefunctions. Red, white and blue denote +1, 0 and -1, respectively. The native plaquettes are displayed without interpolation in dimensionless reduced coordinates, $\mathbf{k} = q_1 \mathbf{b}_1 + q_2 \mathbf{b}_2$, with opposite edges periodically identified. The line at $q_2 = 0$ separates the upper and lower half zones. The occupied-band calculation gives $Z_2 = 1$.

The upper and lower half-zone sums are **-3 and +3**, respectively. Both are odd, so their parities agree at $\nu = 1$, identifying the nontrivial time-reversal-symmetric insulating phase for this calculation. The sampled minimum direct and global gaps are **0.592 eV** and **0.510 eV**.

The local n-field depends on the gauge and logarithm branch; its half-zone parity is the invariant. The reduced-coordinate map shows the original mesh half-zones used for these sums. The separate occupied $\mathbf{C} = \mathbf{0}$ and zero charge-Hall response are consistent with this nontrivial Z_2 result.

Z_2 calculation and plotting guide · Numerical n-field data

3.5 MoS₂: a real-space Γ state

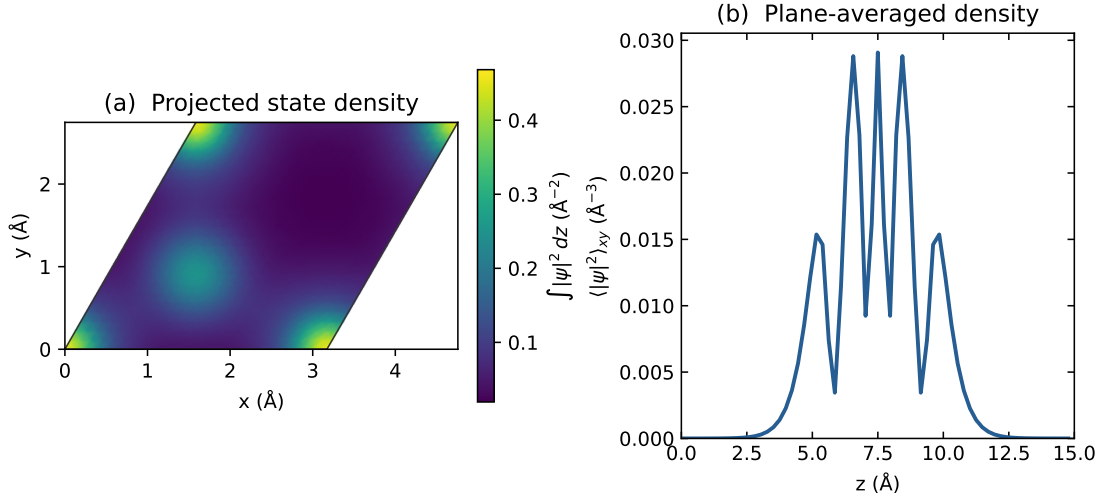


Figure 6. Projected pseudo-wavefunction density of band 18 at Γ and its plane average along z . The in-plane plot uses the Cartesian geometry of the oblique primitive cell. Both spinor components are retained.

The integrated pseudo-density is approximately **0.8258**, consistent with the plane-wave coefficient norm. The PAW augmentation is not part of this plot, so a unit density integral is not imposed. The example illustrates conversion of complex amplitudes into a physical-coordinate density map.

Wavefunction calculation guide

4. Practical use

Begin with the example guide, reproduce the chosen quantity, and then select the sampling and band window appropriate to your own material. Use Cartesian reciprocal-space maps to display lattice symmetry, named k paths for band-resolved quantities, and energy or chemical potential for spectra and transport. State units, represented bands, temperature and broadening in the caption.

For quantitative conclusions, converge the VASP electronic structure, k mesh and relevant band window. The examples show how the methods are used; the supplementary analytic references check formulas and conventions, while validation details document the numerical implementation. Detailed execution records remain alongside the numerical files for reproducibility.

References

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